

COMPUTATIONAL IMAGING

Spring 2024

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Master in Robotics, Graphics and Computer Vision – Universidad de Zaragoza

Instructors



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* Office hours: Check Moodle for info on how each of us will organize them.

Schedule - Overview

- Three parts:
 - Lectures : Wed (1h), Thu (2h)
 - Starting January 24 (Thu)
 - Lab sessions: Mon & Wed (2h)
 - Starting February 5 (Monday)
 - Deadlines throughout the semester.
 - Course project:
 - Developed through the entire semester.
 - Mentoring sessions during lecture hours.

Lectures - Organization

- *When?*
 - Wednesday 10:00 - 10:50
 - Thursday 10:00 - 11:50
- *Where?* Classroom A.07, Ada Byron
- Starting January 24. Check the MRGCV Google Calendar for full schedule.

Resources

Moodle: <https://moodle.unizar.es/add/course/view.php?id=102161>

- General information & bulletin board
- Lecture notes
- Lab scripts
- All submissions related to lab assignments or the course project

* If you are not enrolled but will be following the course, contact us.

Continuous evaluation

You will be graded based on:

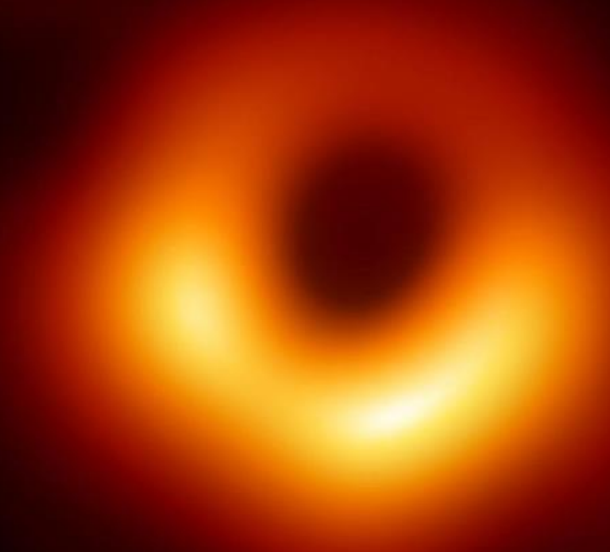
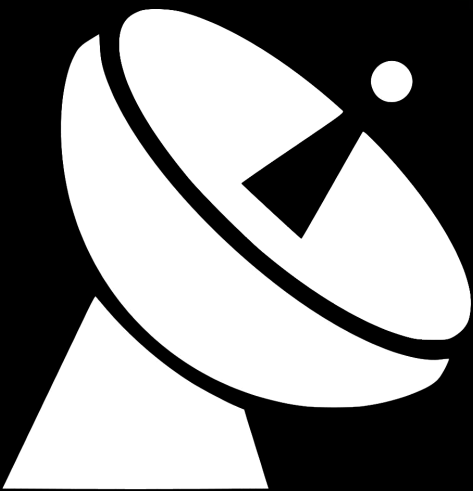
- Lab assignments (6): 40%
 - Course project: 45%
 - Presentations and debates: 15%
-
- Submission deadlines will be posted on the Google Calendar in due time.
 - Submitting all six lab assignments is required. Presenting the course project is required as well.
 - Lab assignments will be done **in pairs**. Grading may be done individually.
 - Course project will be done in **groups of four**. Grading may be done individually.
 - You'll need a grade >4 in all of them, with a weighted average >5

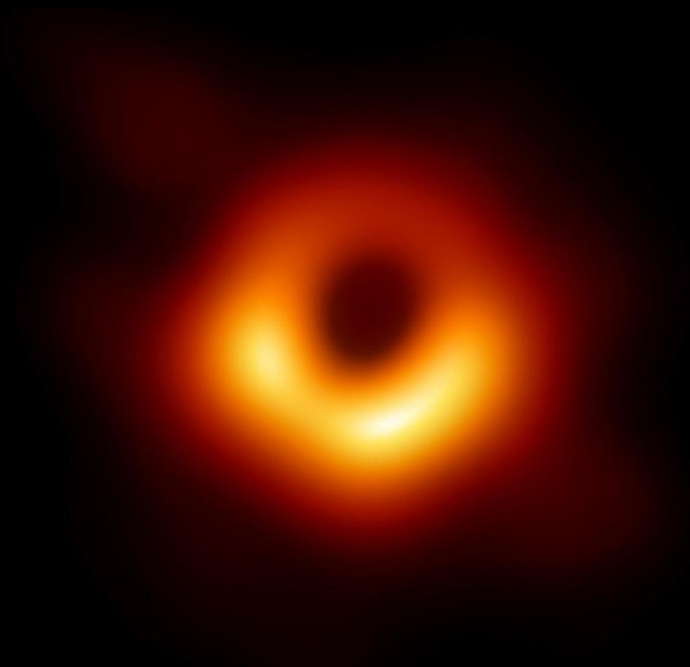
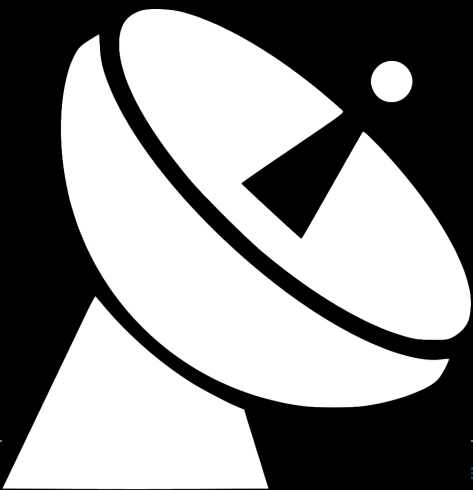
Global evaluation

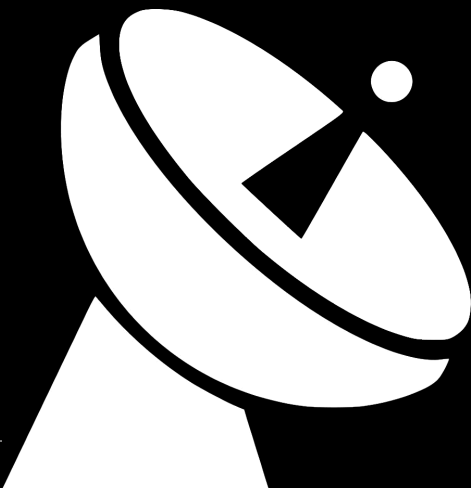
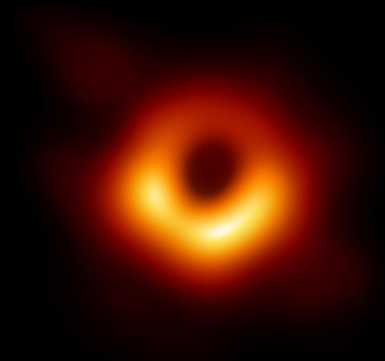
You will be graded based on:

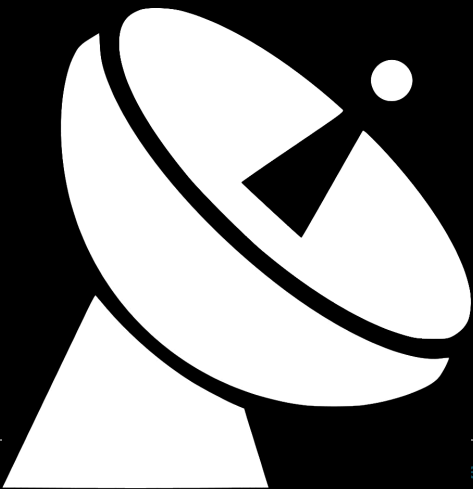
- Global examination: 60%
 - Course project: 25%
 - Presentation: 15%
-
- Global evaluation for all parts:
 - 1st call: **May 23, 2024**
 - 2nd call: **June 25, 2024**
 - The course project under global evaluation may be done in groups or individually.
 - You'll need a grade >4 in all of them, with a weighted average >5

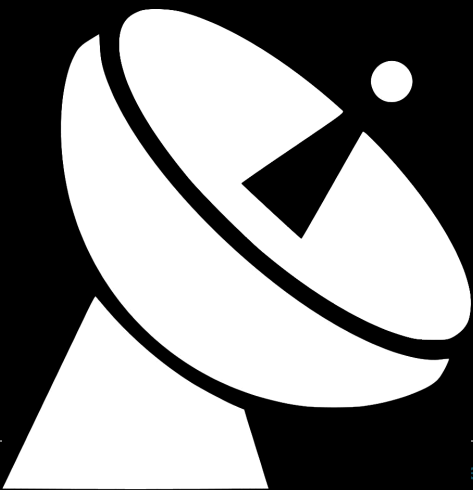
OK, OK. But what is computational imaging?



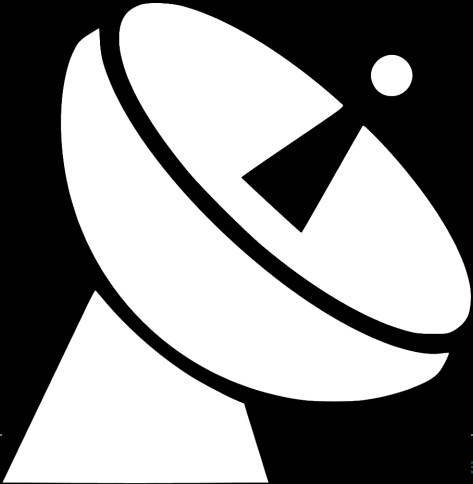








0.00004 as

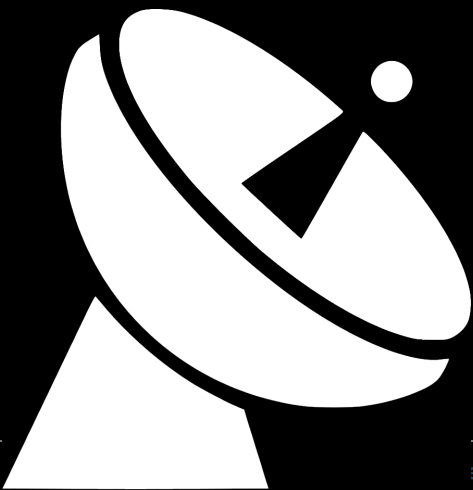


Black Hole Simulation

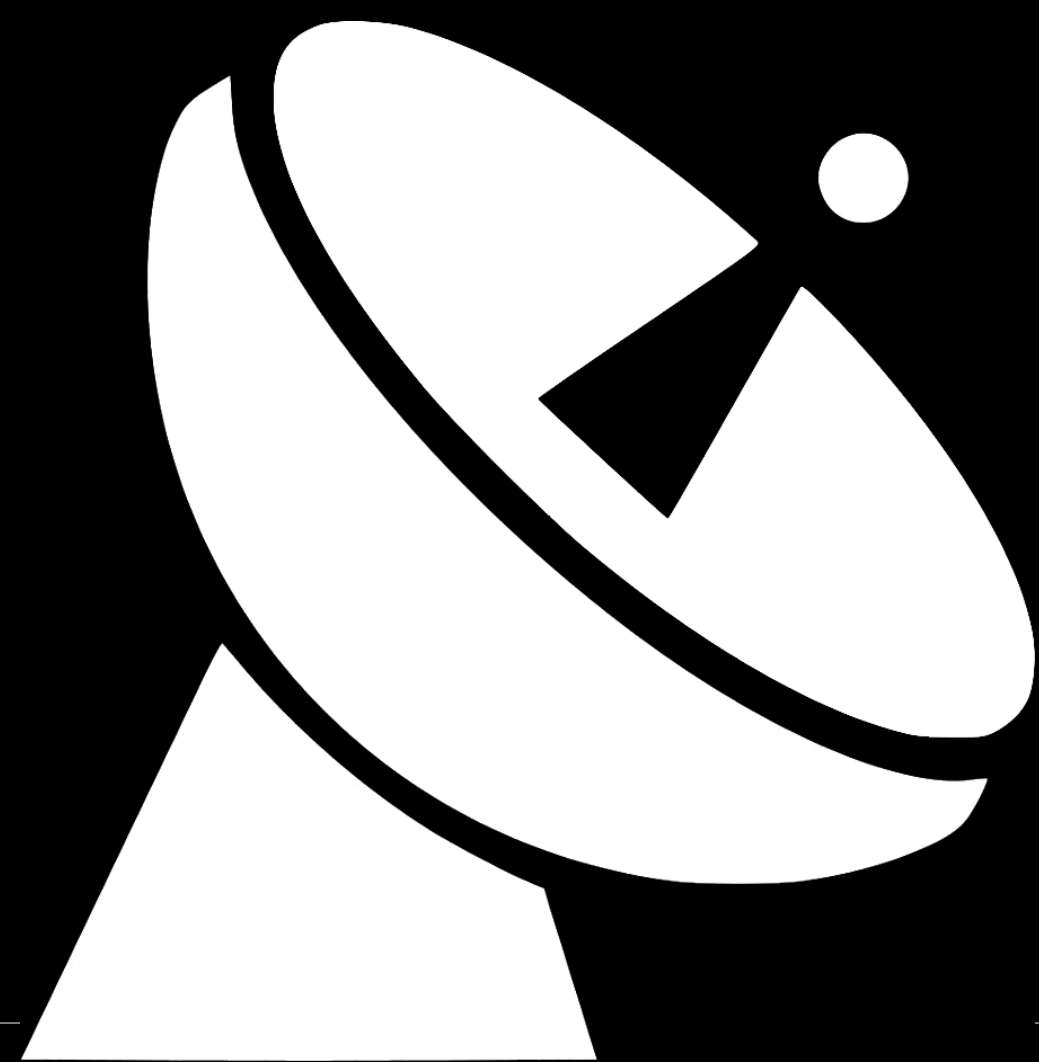
0.00004 as

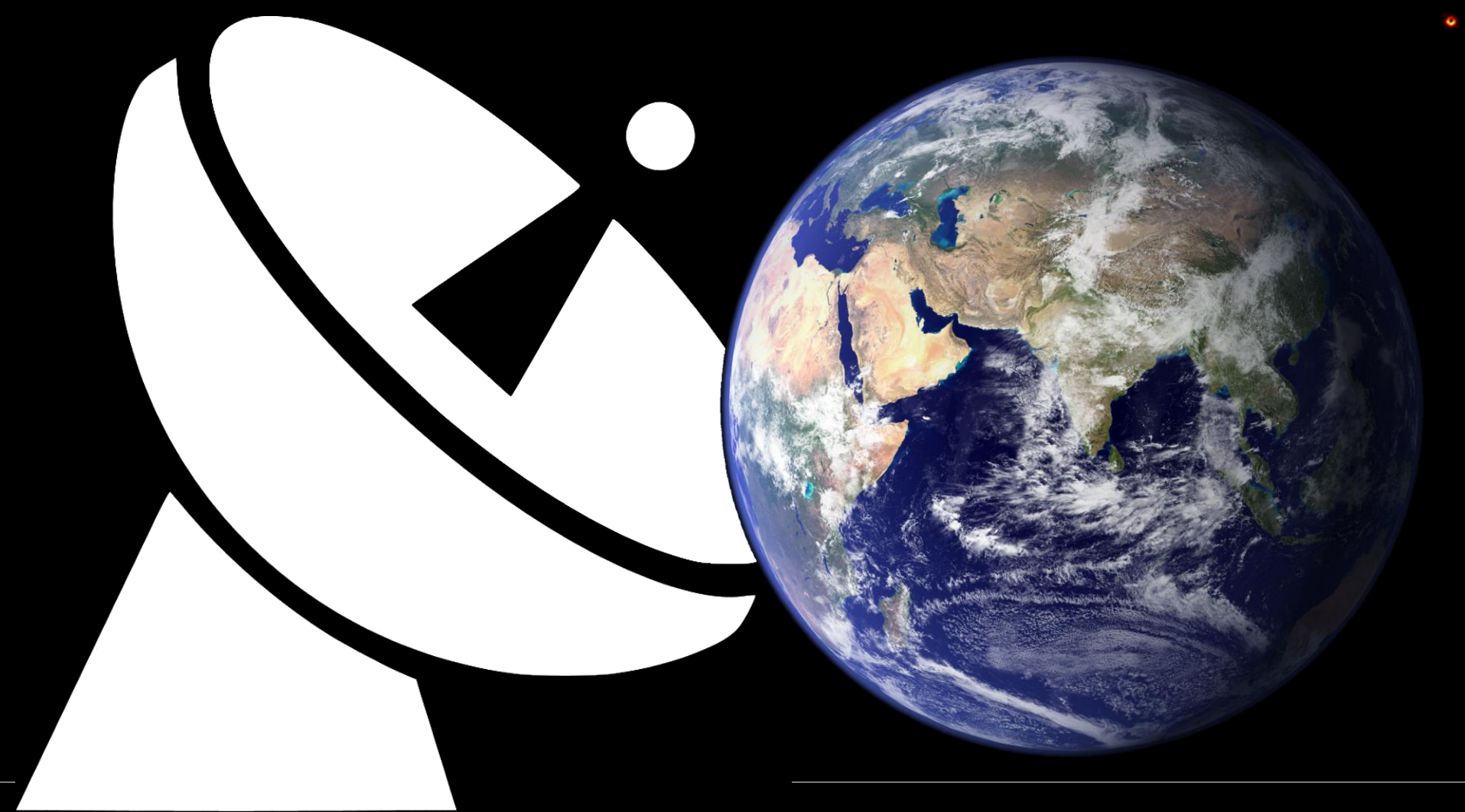
Grain of Sand
in New York



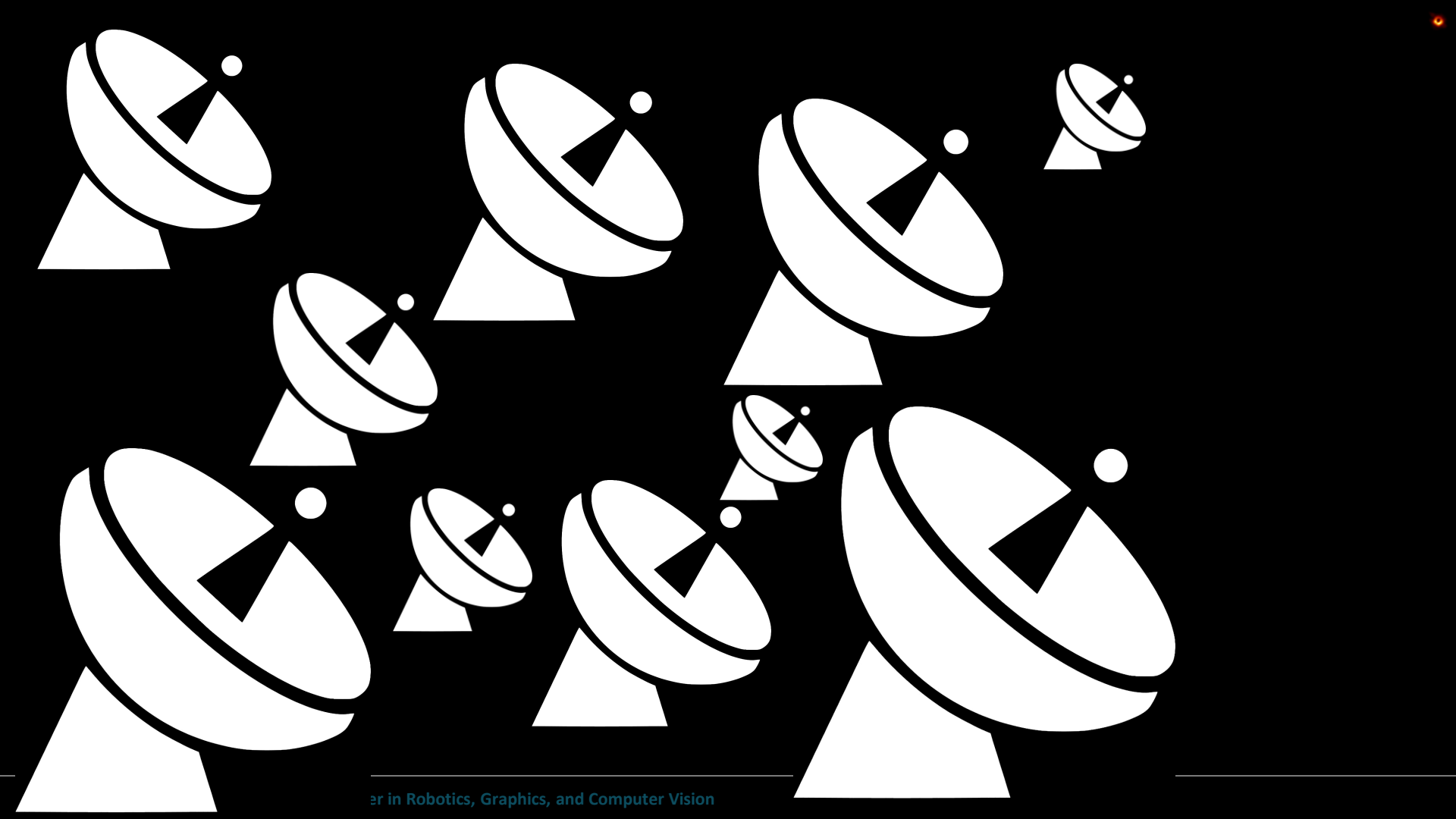


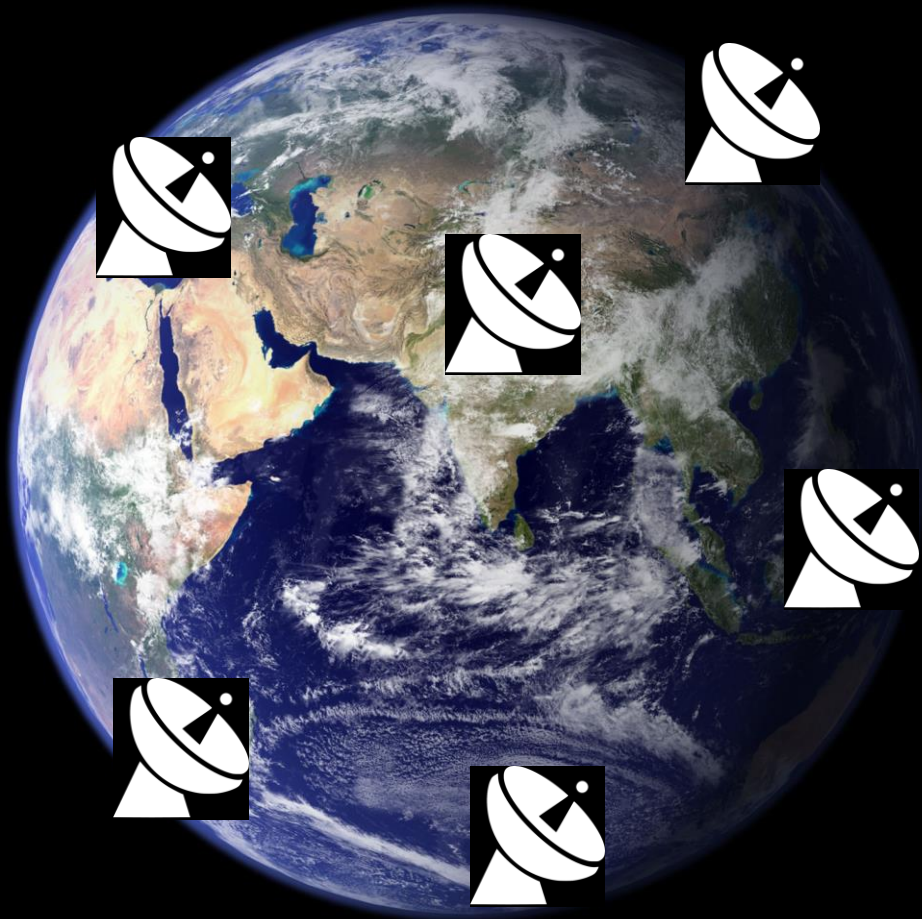


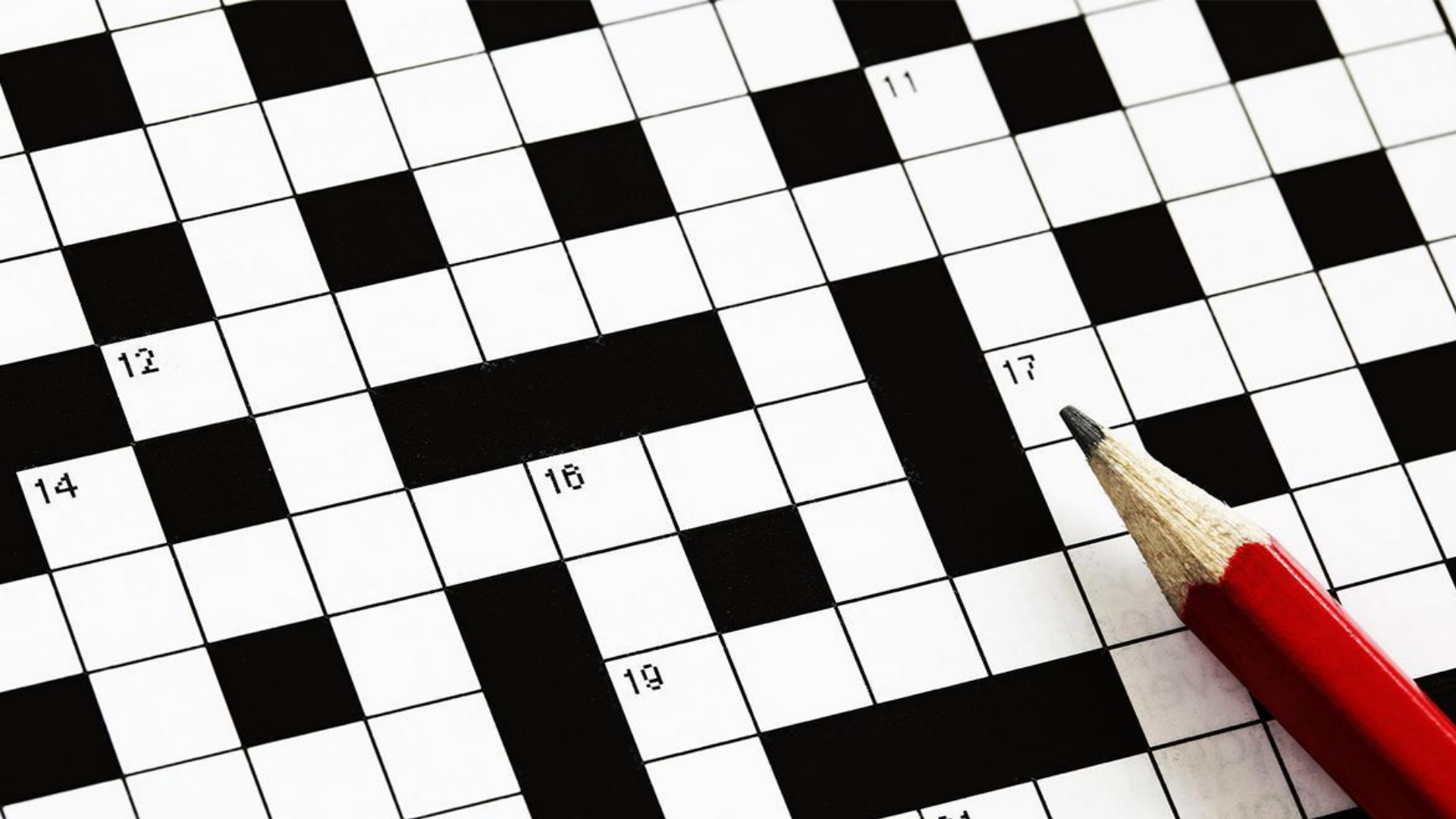




Computational imaging







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Photographs by Wojciech Jarosz



Photographs by Wojciech Jarosz



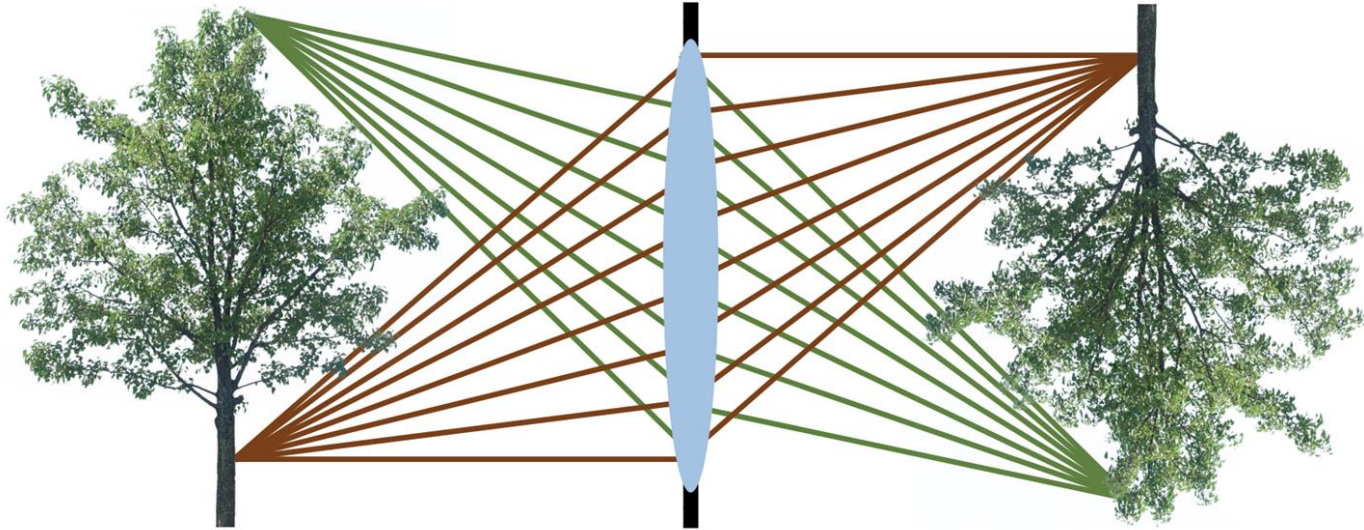
Photographs by Wojciech Jarosz

Course program

- Image formation, digital photography, and image processing
- Mathematical foundations
- Plenoptic sampling, coded apertures, sparse coding
- Light field imaging
- High-dynamic-range imaging
- Computational displays
- Computational illumination
- Hyperspectral and polarization imaging
- Transient imaging
- *Course project presentations*

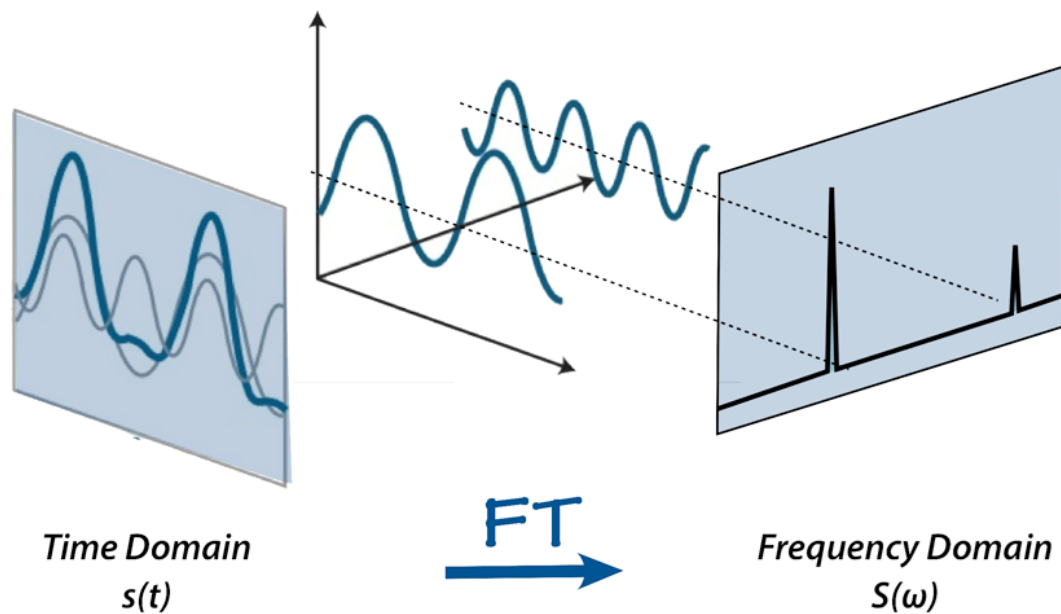
Course program

Image formation



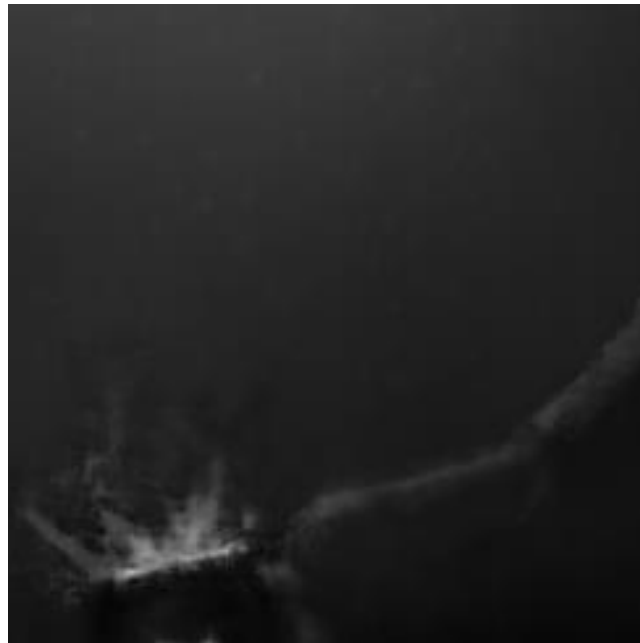
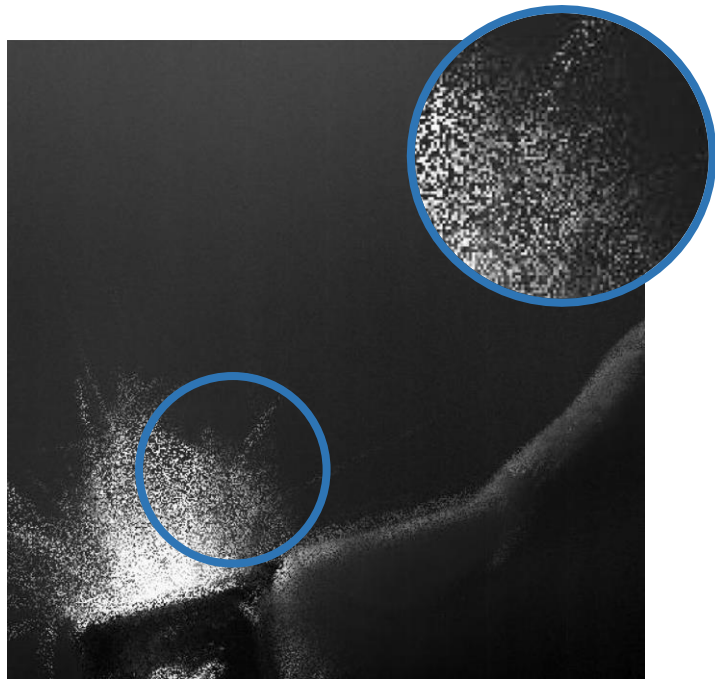
Course program

Mathematical foundations



Course program

Plenoptic sampling, coded apertures, sparse coding



Course program

Light Field imaging



Course program

High-dynamic-range (HDR) Imaging



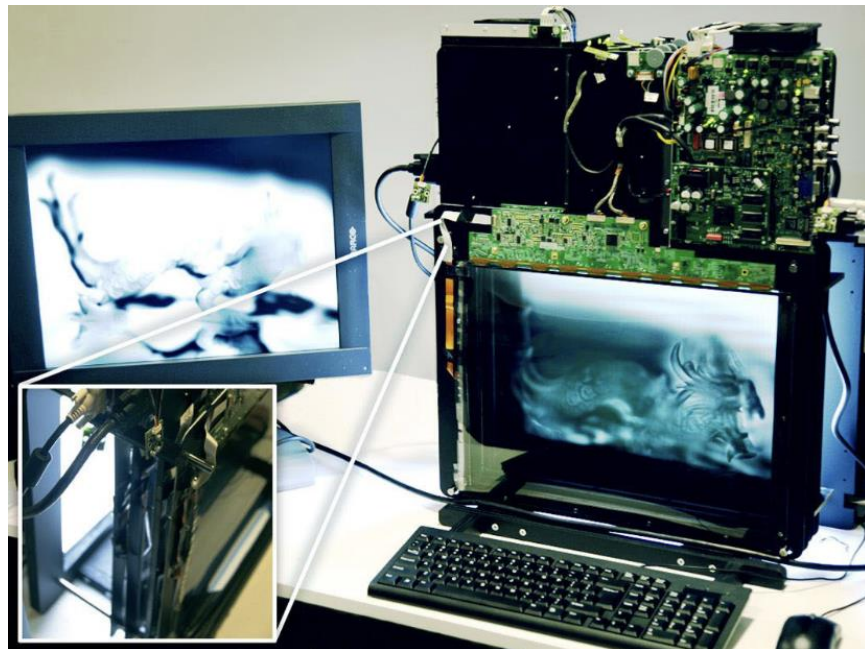
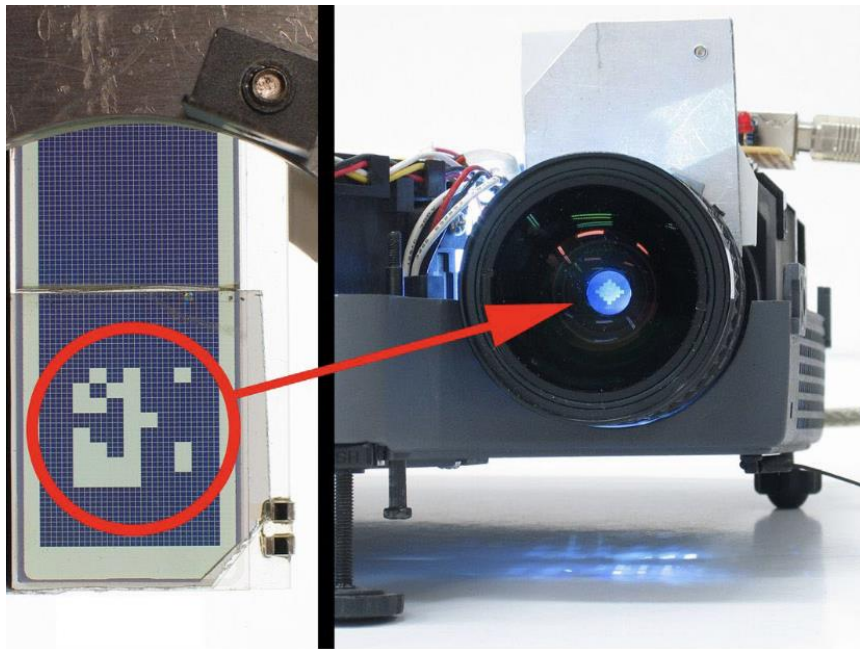
Course program

High-dynamic-range (HDR) Imaging



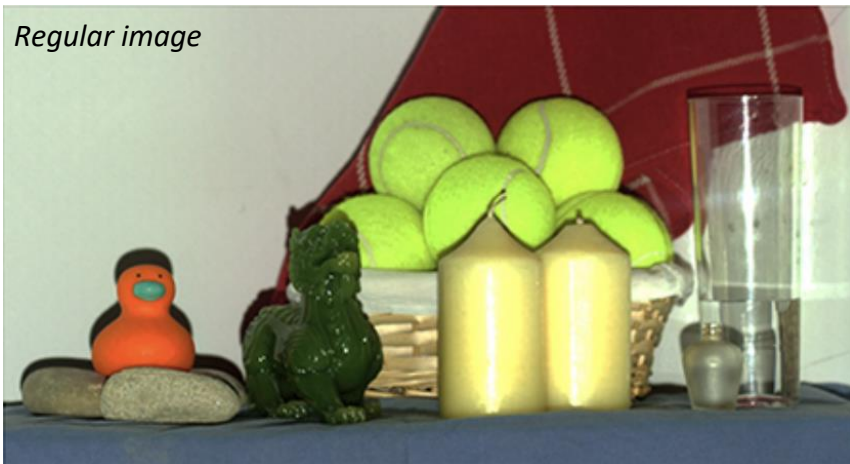
Course program

Computational displays



Course program

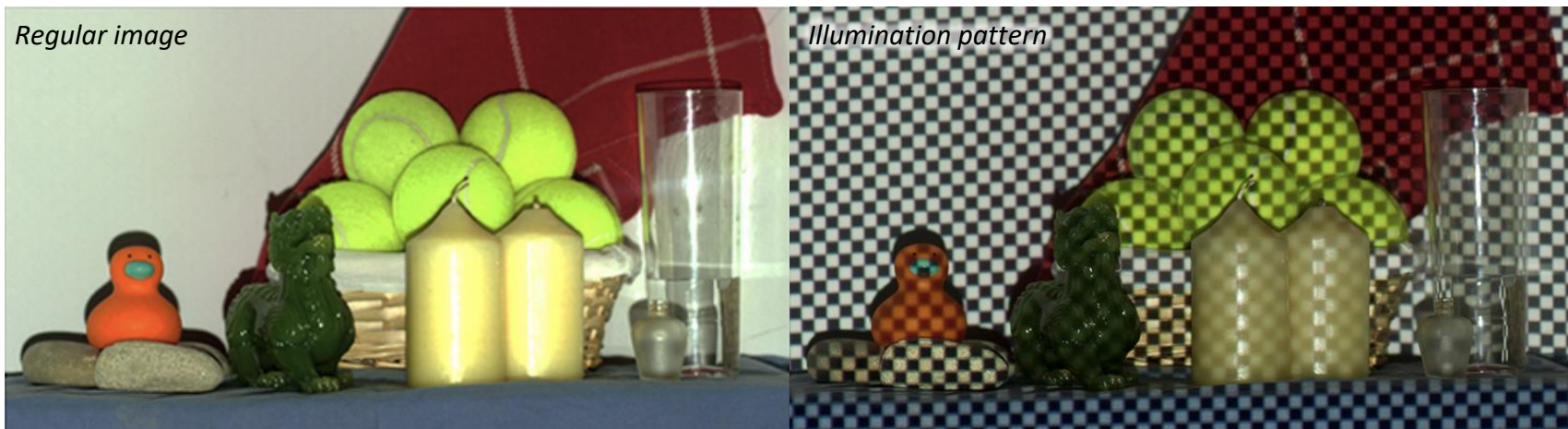
Computational illumination



[Naik et al. 2006]

Course program

Computational illumination



[Naik et al. 2006]

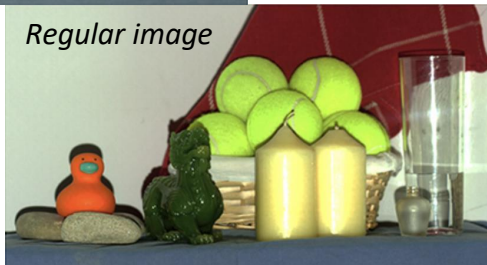
Course program

Computational illumination

Direct light



Regular image



[Naik et al. 2006]

Course program

Computational illumination

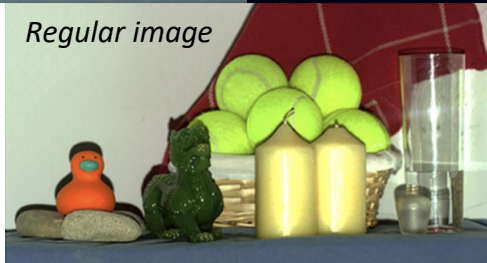
Direct light



Indirect light



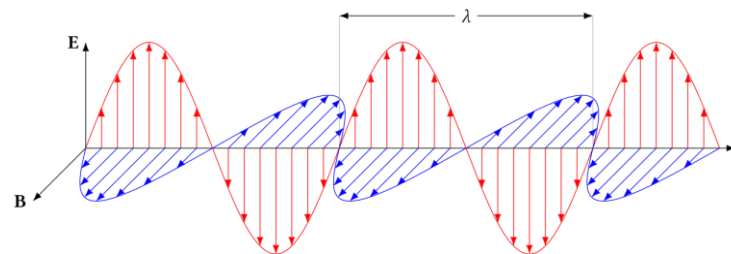
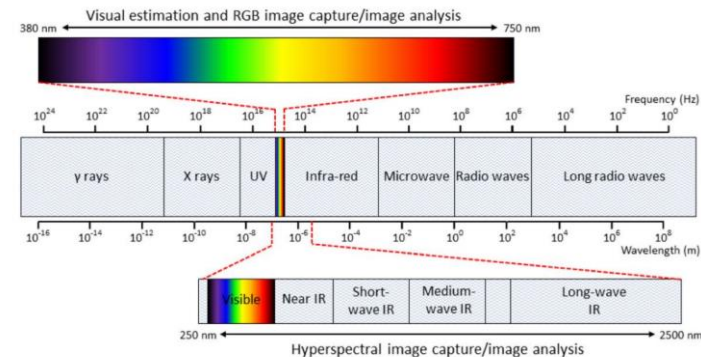
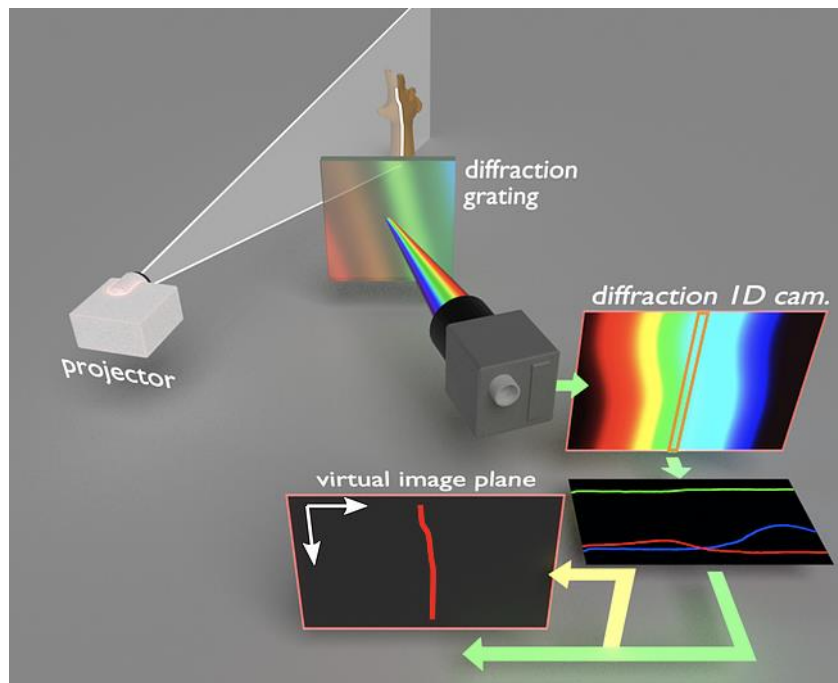
Regular image



[Naik et al. 2006]

Course program

Hyperspectral Imaging & Polarization



Course program

Transient imaging

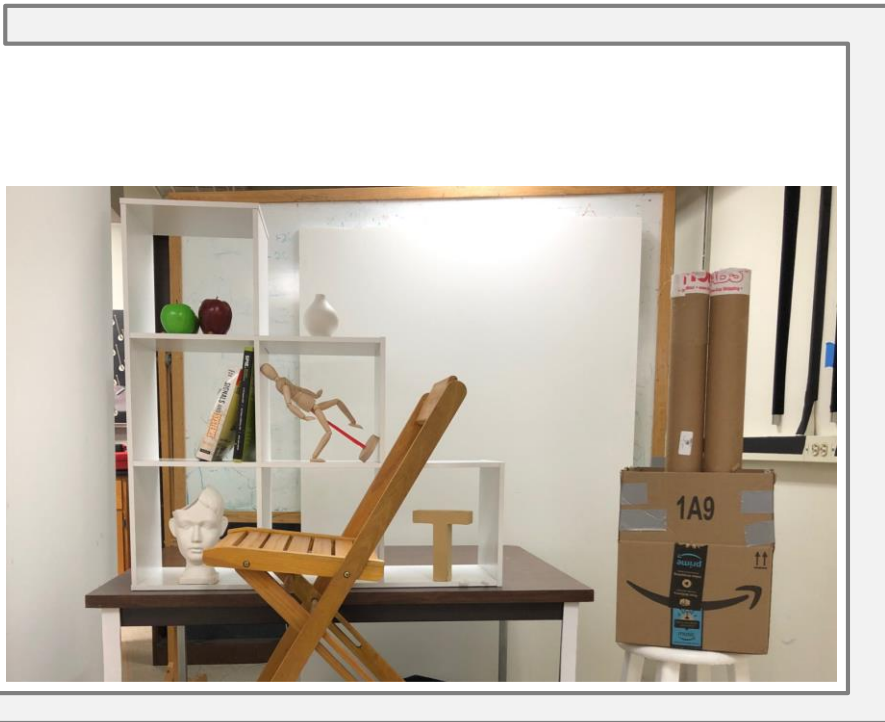


Femto-photography [Velten et al. 2013]

Course program



Transient imaging





Phasor fields [Liu et al., 2019]



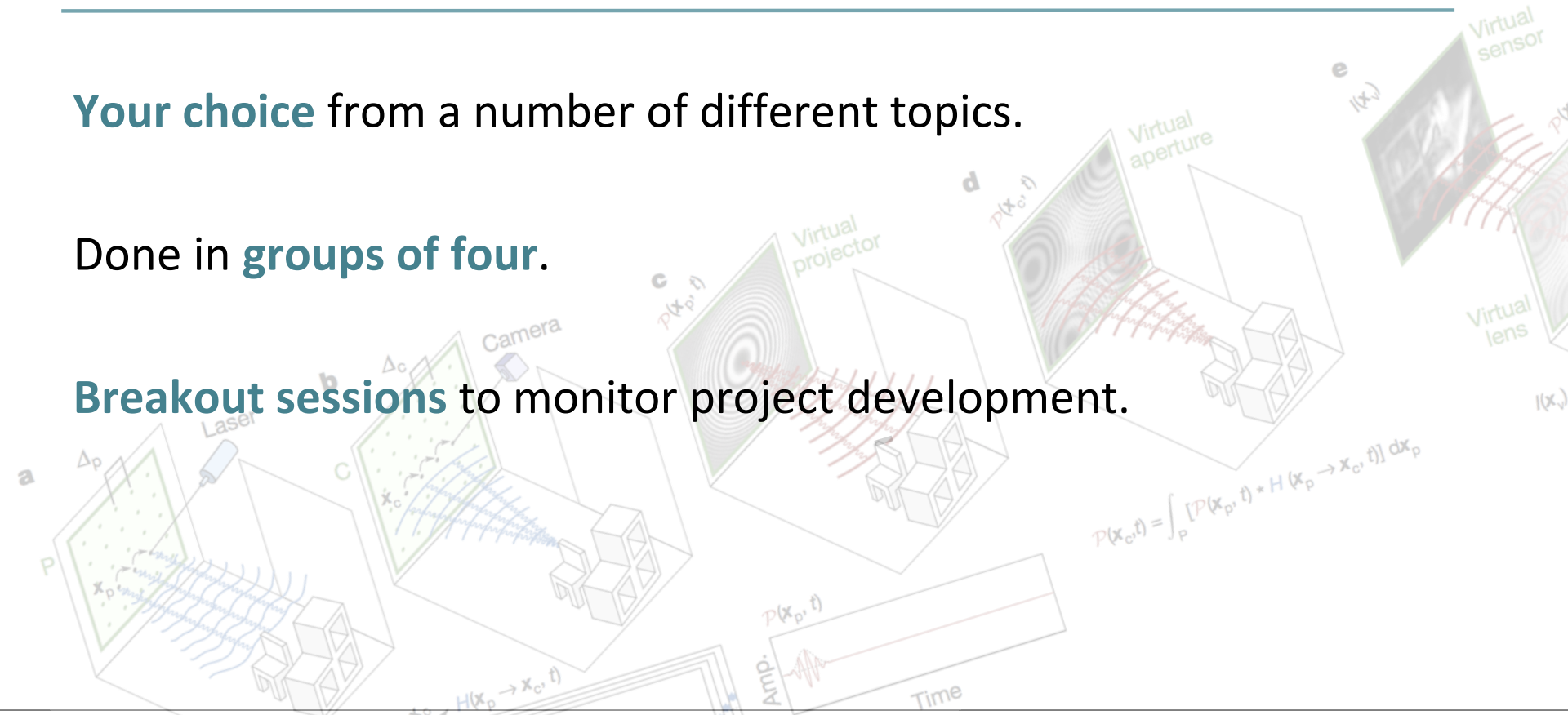
Phasor fields [Liu et al., 2019]

Course project

Your choice from a number of different topics.

Done in **groups of four**.

Breakout sessions to monitor project development.



Course project – Tentative timeline

- Image formation, digital photography, and image processing
- Mathematical foundations
- Plenoptic sampling, coded apertures, sparse coding *Course project introduction*
- Light field imaging *Course project development*
- High-dynamic-range imaging
- Computational displays *Course project development*
- Computational illumination
- Hyperspectral and polarization imaging *Course project development*
- Transient imaging *Course project development*
-
- May 15-16 (tentative)** *Course project presentations*

See you tomorrow!

